

ESE5180: Lab 0 Zephyr

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GitHub Repository URL: <https://github.com/Bowen-wang-sps/ese5180-lab0-zephyr.git>

(3.1) Build and Flash using only west commands and not the GUI. Show the terminal prints by embedding ascreenshot in your README.m

```
PS C:\Users\sps\Desktop\code\ese5180\ese5180-lab0-zephyr> west build --build-dir l1_e2/build l1_e2 --pristine --board nrf7002dk/nrf5340/cpuapp/ns --sysbuild -DBOARD_ROOT=l1_e2
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found suitable version "3.12.4", minimum required is "3.12") found components: Interpreter
-- Cache files will be written to: C:/ncs/v3.4.0/zephyr/.cache
-- Found Python3: C:/ncs/toolchains/dcbdc366a1/opt/bin/python.exe (found version "3.12.4") found components: Interpreter
CMake Warning (dev) at C:/ncs/v3.4.0/nrf/modules/trusted-firmware-m/tfm_boards/nrf5340_cpuapp/CMakeLists.txt:21 (install):
  Policy CMP0177 is not set: install() DESTINATION paths are normalized. Run
  "cmake --help-policy CMP0177" for policy details. Use the cmake_policy
  command to set the policy and suppress this warning.
This warning is for project developers. Use -Wno-dev to suppress it.

-- Configuring done (2.1s)
-- Generating done (0.3s)
CMake Warning:
  Manually-specified variables were not used by the project:

  QCBOR_PATH
  TFM_CREATE_SIGNING_LAYOUT_OBJECTS
  T_COSE_PATH

-- Build files have been written to: C:/Users/sps/Desktop/code/ese5180/ese5180-lab0-zephyr/l1_e2/build/l1_e2/tfm
[193/195] Linking C executable bin\tfm_s.axf
Memory region      Used Size  Region Size  %age Used
  FLASH:           81364 B      256 KB      31.04%
    RAM:          45960 B      256 KB      17.53%
[195/195] Linking C static library secure_fw\libtfm_s_veneers.a

[17/259] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[259/259] Linking C executable zephyr\zephyr.elf
Memory region      Used Size  Region Size  %age Used
  FLASH:           22576 B       704 KB       3.13%
    RAM:           4600 B       192 KB       2.34%
  IDT_LIST:          0 B        32 KB       0.00%
Generating files from C:/Users/sps/Desktop/code/ese5180/ese5180-lab0-zephyr/l1_e2/build/l1_e2/zephyr/zephyr.elf for board: nrf7002dk/nrf5340/cpuapp/ns
[8/8] Completed 'l1_e2'
```

```

PS C:\Users\sps\Desktop\code\ese5180\ese5180-lab0-zephyr> west flash -d l1_e2/build --dev-id 1050753830 --erase
-- west flash: rebuilding
[0/16] Performing build step for 'tfm'ninja: no work to do.

[2/3] Performing install step for 'tfm'
-- Install configuration: "MinSizeRel"
----- Installing platform NS -----
[3/3] Completed 'tfm'

[3/3] Completed 'l1_e2'
-- west flash: using runner nrfutil
-- runners.nrfutil: mass erase requested
-- runners.nrfutil: reset after flashing requested
-- runners.nrfutil: Flashing file: C:\Users\sps\Desktop\code\ese5180\ese5180-lab0-zephyr\l1_e2\build\l1_e2\zephyr\tfm_merged.hex
-- runners.nrfutil: Connecting to probe
-- runners.nrfutil: Erasing non-volatile memory (ERASEALL)
-- runners.nrfutil: Programming image
-- runners.nrfutil: Verifying image
-- runners.nrfutil: Reset
-- runners.nrfutil: Board(s) with serial number(s) 1050753830 flashed successfully.

```

(6.1) Take screenshots of console output for both builds: ● CONFIG_SUM_PRINT=y → result printed with printk(). ● CONFIG_SUM_LOG=y → result printed with the Logger (include hexdump).

```

*** Booting nRF Connect SDK v3.4.0-99553055607b ***
*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
2 + 3 = 5
|

```

```

*** Using Zephyr OS v4.4.0-bf801e4e3d19 ***
[00:00:00.393,127] <inf> sum_log: result: 5
[00:00:00.393,127] <wrn> sum_log: warning
[00:00:00.393,157] <err> sum_log: error
[00:00:00.393,188] <inf> sum_log: inputs
                        02 00 00 00 03 00 00 00
|.....

```

(7.2) Take screenshots of the testing outputs (laptop) in the terminal.

```

INFO - Using Ninja..
INFO - Zephyr version: ncs-v3.4.0
INFO - Using 'zephyr/gnu' toolchain variant.
INFO - Building initial testsuite list...
INFO - Built testsuite list in 0.00 seconds
INFO - Writing JSON report C:\twister-out\testplan.json
INFO - Adding tasks to the queue...
INFO - Added initial list of jobs to queue
INFO - 1/1 qemu_cortex_m3/ti_lm3s6965 sum_log.test PASSED (qemu 10.377s <zephyr/gnu>)

INFO - 1 test scenarios (1 configurations) selected, 0 configurations filtered (0 by static filter, 0 at runtime).
INFO - 1 of 1 executed test configurations passed (100.00%), 0 built (not run), 0 failed, 0 errored, with no warnings in 35.25 seconds.
INFO - 3 of 3 executed test cases passed (100.00%) on 1 out of total 1514 platforms (0.07%).
INFO - 1 test configurations executed on platforms, 0 test configurations were only built.
INFO - Saving reports...
INFO - Writing JSON report C:\twister-out\twister.json
INFO - Writing xunit report C:\twister-out\twister.xml...
INFO - Writing xunit report C:\twister-out\twister_report.xml...
INFO - Run completed

```

(8.1) Print out the temperature. Take screenshots of your logging output. Commit your Zephyr application to your GitHub repository